

Indian Institute of Technology Hyderabad

Signal Processing, Fall 2026

Quiz 1, 17.08.2026, 10 points

1. Discrete-time Fourier Transform (DTFT):
 - (a) Recall the discrete-time system that computes the average of the *current* sample and the *sample before its previous* sample. Plot its DTFT magnitude response for $0 \leq \omega < 2\pi$. (1)
 - (b) The DTFT phase response is defined as the angle of the DTFT as a function of ω . Plot the DTFT phase response of the *first-order differencing system* discussed in class for $0 \leq \omega < 2\pi$. (2)
2. Given an example of a signal whose z-transform exists but whose DTFT does not. Find the *region of convergence* of the z-transform. Justify your answer. (2)
3. A linear system is said to satisfy two properties: (1) additivity and (2) scaling or homogeneity. Why do we need both? Does additivity not automatically imply scaling? Justify your answer. (2)
4. Check if the following systems are *linear*. Justify your answer. (3)
 - (a) $y[n] = T\{x[n]\} = ax[n] + b$, where a and b are constants.
 - (b) $y[n] = T\{x[n]\} = x[Mn]$, where M is a positive integer.
 - (c) $y[n] = T\{x[n]\} = x[n-1] + x[n+1]$.